

Question number	Answer	Additional guidance	Mark
8(a)	A description that includes three of the following points: • a muscle fibre is a single cell (1) • a skeletal muscle cell which is multinucleate(1) • striated appearance due to repeating bands of actin and myosin (1) • actin and myosin alternate to form the sarcolemma (1)	Accept suitably labelled diagram. Accept reference to sarcomere.	(3)

Question number	Answer	Additional guidance	Mark
8(b)	An explanation that includes three of the following points: • because myokines are released in response to muscle contraction (1) • because they travel in blood and have stated endocrine effects on liver (1) • therefore myokines have direct effect on muscle itself – autocrine (1) • so myokines regulate muscle angiogenesis / adipogenesis / innervation (1)	e.g. Effects on metabolism, stimulation of respiration, increased fat metabolism, glycogenesis, and regulation of homeostasis. credit reference to binding to receptors on cell surface credit activating transcription factors	(3)

Question number	Answer	Additional guidance	Mark
8(c)	An explanation that includes four of the following points: • because loss of muscle mass makes movement more difficult • because reduced muscle strength makes moving from the seated position a challenge • because of reduction in the number of satellite cells means that muscle repair is reduced • reduction in speed of action/ response due to less muscle innervation	Responses must be a clear comparison either in one sentence or in adjacent sentences. Accept correct reference to sarcopaenia Accept fewer muscle fibres (and an increase in fat and connective tissues) Accept correct reference to slow and fast twitch muscle fibres. – decrease in fast twitch fibres or an increase in slow twitch fibres	(4)

Question number	Answer	Additional guidance	Mark
8(d)	An explanation that includes three of the following points: • this causes change in shape of tropomyosin freeing the myosin heads • myosin heads bind with actin forming cross bridges • ATP breakdown releases energy which allows myosin to pull the actin inwards, causing shortening.	accept power stroke	(3)

Question number	Answer	Additional guidance	Mark
8(e)	A description that includes two of the following points: • names of two or more 'high energy phosphate' compounds involved in muscle contraction (1) • hydrolysis of phosphate bond in ATP to release energy (1) • creatine phosphate provides phosphate to ADP to produce more ATP to continue contraction (1)	ATP, ADP, creatine phosphate (phosphocreatine) ATP in muscle releases energy for myosin to pull the actin molecule	(2)

Question number	Answer	Additional guidance	Mark
8(f)	An answer that includes three of the following points: • leads to a reduction in contractile force at the joint • causes a change in tendon materials • causes a reduced type of movement at the joint • stated implications for the individual for effectively reacting to avoid a slip or fall	Accept results in loss of collagen / increase in elastin/reduction in extracellular water content Accept tendon would stretch less on muscle contraction / sarcomere shortening	(3)

Question number	Answer	Additional guidance	Mark
8(g)	An answer that includes two of the following points: • acts as a transcription factor in muscle growth at gene level (1) • inhibits myogenesis / muscle growth / cell differentiation (1) • promotes muscle wasting (1)	Perhaps: activates genes involved in myogenesis / for proteases that break down muscle protein	(2)

